



PRINCIPLES IN PRACTICE

SETTING OSPF LINK WEIGHTS

Our discussion of link-state routing has implicitly assumed that link weights are set, a routing algorithm such as OSPF is run, and traffic flows according to the routing tables computed by the LS algorithm. In terms of cause and effect, the link weights are given (i.e., they come first) and result (via Dijkstra's algorithm) in routing paths that minimize overall cost. In this viewpoint, link weights reflect the cost of using a link (e.g., if link weights are inversely proportional to capacity, then the use of high-capacity links would have smaller weights and thus be more attractive from a routing standpoint) and Dijkstra's algorithm serves to minimize overall cost.

In practice, the cause and effect relationship between link weights and routing paths may be reversed, with network operators configuring link weights in order to obtain routing paths that achieve certain traffic engineering goals [Fortz 2000, Fortz 2002]. For example, suppose a network operator has an estimate of traffic flow entering the network at each ingress point and destined for each egress point. The operator may then want to put in place a specific routing of ingress-to-egress flows that minimizes the maximum utilization over all of the network's links. But with a routing algorithm such as OSPF, the operator's main "knobs" for tuning the routing of flows through the network are the link weights. Thus, in order to achieve the goal of minimizing the maximum link utilization, the operator must find the set of link weights that achieves this goal. This is a reversal of the cause and effect relationship—the desired routing of flows is known, and the OSPF link weights must be found such that the OSPF routing algorithm results in this desired routing of flows.

OSPF is a relatively complex protocol, and our coverage here has been necessarily brief; [Huitema 1998; Moy 1998; RFC 2328] provide additional details.

4.6.3 Inter-AS Routing: BGP

We just learned how ISPs use RIP and OSPF to determine optimal paths for source-destination pairs that are internal to the same AS. Let's now examine how paths are determined for source-destination pairs that span multiple ASs. The **Border Gateway Protocol** version 4, specified in RFC 4271 (see also [RFC 4274]), is the *de facto* standard inter-AS routing protocol in today's Internet. It is commonly referred to as BGP4 or simply as **BGP**. As an inter-AS routing protocol (see Section 4.5.3), BGP provides each AS a means to

1. Obtain subnet reachability information from neighboring ASs.
2. Propagate the reachability information to all routers internal to the AS.
3. Determine "good" routes to subnets based on the reachability information and on AS policy.

Most importantly, BGP allows each subnet to advertise its existence to the rest of the Internet. A subnet screams “I exist and I am here,” and BGP makes sure that all the ASs in the Internet know about the subnet and how to get there. If it weren’t for BGP, each subnet would be isolated—alone and unknown by the rest of the Internet.

BGP Basics

BGP is extremely complex; entire books have been devoted to the subject and many issues are still not well understood [Yannuzzi 2005]. Furthermore, even after having read the books and RFCs, you may find it difficult to fully master BGP without having practiced BGP for many months (if not years) as a designer or administrator of an upper-tier ISP. Nevertheless, because BGP is an absolutely critical protocol for the Internet—in essence, it is the protocol that glues the whole thing together—we need to acquire at least a rudimentary understanding of how it works. We begin by describing how BGP might work in the context of the simple example network we studied earlier in Figure 4.32. In this description, we build on our discussion of hierarchical routing in Section 4.5.3; we encourage you to review that material.

In BGP, pairs of routers exchange routing information over semipermanent TCP connections using port 179. The semi-permanent TCP connections for the network in Figure 4.32 are shown in Figure 4.40. There is typically one such BGP TCP connection for each link that directly connects two routers in two different ASs; thus, in Figure 4.40, there is a TCP connection between gateway routers 3a and 1c and another TCP connection between gateway routers 1b and 2a. There are also semipermanent BGP TCP connections between routers within an AS. In particular, Figure 4.40 displays a common configuration of one TCP connection for each pair of routers internal to an AS, creating a mesh of TCP connections within each AS. For each TCP connection, the two routers at the end of the connection are called **BGP peers**, and the TCP connection along with all the BGP messages sent over the

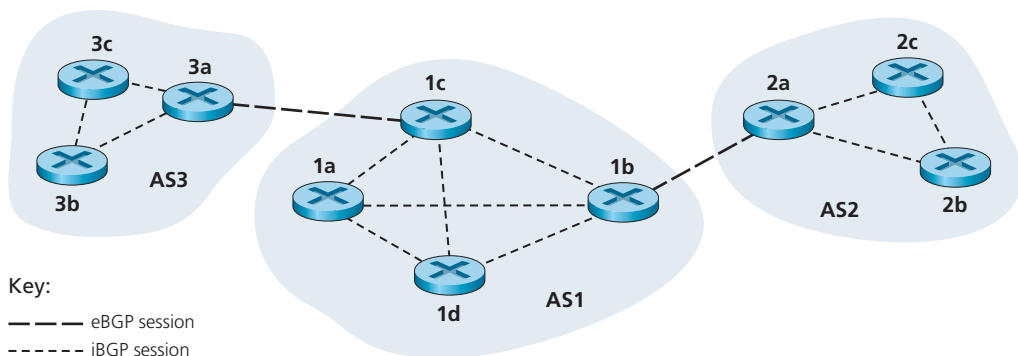


Figure 4.40 ♦ eBGP and iBGP sessions



PRINCIPLES IN PRACTICE

OBTAINING INTERNET PRESENCE: PUTTING THE PUZZLE TOGETHER

Suppose you have just created a small that has a number of servers, including a public Web server that describes your company's products and services, a mail server from which your employees obtain their email messages, and a DNS server. Naturally, you would like the entire world to be able to surf your Web site in order to learn about your exciting products and services. Moreover, you would like your employees to be able to send and receive email to potential customers throughout the world.

To meet these goals, you first need to obtain Internet connectivity, which is done by contracting with, and connecting to, a local ISP. Your company will have a gateway router, which will be connected to a router in your local ISP. This connection might be a DSL connection through the existing telephone infrastructure, a leased line to the ISP's router, or one of the many other access solutions described in Chapter 1. Your local ISP will also provide you with an IP address range, e.g., a /24 address range consisting of 256 addresses. Once you have your physical connectivity and your IP address range, you will assign one of the IP addresses (in your address range) to your Web server, one to your mail server, one to your DNS server, one to your gateway router, and other IP addresses to other servers and networking devices in your company's network.

In addition to contracting with an ISP, you will also need to contract with an Internet registrar to obtain a domain name for your company, as described in Chapter 2. For example, if your company's name is, say, Xanadu Inc., you will naturally try to obtain the domain name `xanadu.com`. Your company must also obtain presence in the DNS system. Specifically, because outsiders will want to contact your DNS server to obtain the IP addresses of your servers, you will also need to provide your registrar with the IP address of your DNS server. Your registrar will then put an entry for your DNS server (domain name and corresponding IP address) in the `.com` top-level-domain servers, as described in Chapter 2. After this step is completed, any user who knows your domain name (e.g., `xanadu.com`) will be able to obtain the IP address of your DNS server via the DNS system.

So that people can discover the IP addresses of your Web server, in your DNS server you will need to include entries that map the host name of your Web server (e.g., `www.xanadu.com`) to its IP address. You will want to have similar entries for other publicly available servers in your company, including your mail server. In this manner, if Alice wants to browse your Web server, the DNS system will contact your DNS server, find the IP address of your Web server, and give it to Alice. Alice can then establish a TCP connection directly with your Web server.

However, there still remains one other necessary and crucial step to allow outsiders from around the world access your Web server. Consider what happens when Alice, who knows the IP address of your Web server, sends an IP datagram (e.g., a TCP SYN segment) to that IP address. This datagram will be routed through the Internet, visiting a series of routers in many different ASes, and eventually reach your Web server. When

any one of the routers receives the datagram, it is going to look for an entry in its forwarding table to determine on which outgoing port it should forward the datagram. Therefore, each of the routers needs to know about the existence of your company's /24 prefix (or some aggregate entry). How does a router become aware of your company's prefix? As we have just seen, it becomes aware of it from BGP! Specifically, when your company contracts with a local ISP and gets assigned a prefix (i.e., an address range), your local ISP will use BGP to advertise this prefix to the ISPs to which it connects. Those ISPs will then, in turn, use BGP to propagate the advertisement. Eventually, all Internet routers will know about your prefix (or about some aggregate that includes your prefix) and thus be able to appropriately forward datagrams destined to your Web and mail servers.

connection is called a **BGP session**. Furthermore, a BGP session that spans two ASs is called an **external BGP (eBGP) session**, and a BGP session between routers in the same AS is called an **internal BGP (iBGP) session**. In Figure 4.40, the eBGP sessions are shown with the long dashes; the iBGP sessions are shown with the short dashes. Note that BGP session lines in Figure 4.40 do not always correspond to the physical links in Figure 4.32.

BGP allows each AS to learn which destinations are reachable via its neighboring ASs. In BGP, destinations are not hosts but instead are CIDRized **prefixes**, with each prefix representing a subnet or a collection of subnets. Thus, for example, suppose there are four subnets attached to AS2: 138.16.64/24, 138.16.65/24, 138.16.66/24, and 138.16.67/24. Then AS2 could aggregate the prefixes for these four subnets and use BGP to advertise the single prefix to 138.16.64/22 to AS1. As another example, suppose that only the first three of those four subnets are in AS2 and the fourth subnet, 138.16.67/24, is in AS3. Then, as described in the Principles and Practice in Section 4.4.2, because routers use longest-prefix matching for forwarding datagrams, AS3 could advertise to AS1 the more specific prefix 138.16.67/24 and AS2 could *still* advertise to AS1 the aggregated prefix 138.16.64/22.

Let's now examine how BGP would distribute prefix reachability information over the BGP sessions shown in Figure 4.40. As you might expect, using the eBGP session between the gateway routers 3a and 1c, AS3 sends AS1 the list of prefixes that are reachable from AS3; and AS1 sends AS3 the list of prefixes that are reachable from AS1. Similarly, AS1 and AS2 exchange prefix reachability information through their gateway routers 1b and 2a. Also as you may expect, when a gateway router (in any AS) receives eBGP-learned prefixes, the gateway router uses its iBGP sessions to distribute the prefixes to the other routers in the AS. Thus, all the routers in AS1 learn about AS3 prefixes, including the gateway router 1b. The gateway router 1b (in AS1) can therefore re-advertise AS3's prefixes to AS2. When a router (gateway or not) learns about a new prefix, it creates an entry for the prefix in its forwarding table, as described in Section 4.5.3.

Path Attributes and BGP Routes

Having now a preliminary understanding of BGP, let's get a little deeper into it (while still brushing some of the less important details under the rug!). In BGP, an autonomous system is identified by its globally unique **autonomous system number (ASN)** [RFC 1930]. (Technically, not every AS has an ASN. In particular, a so-called stub AS that carries only traffic for which it is a source or destination will not typically have an ASN; we ignore this technicality in our discussion in order to better see the forest for the trees.) AS numbers, like IP addresses, are assigned by ICANN regional registries [ICANN 2012].

When a router advertises a prefix across a BGP session, it includes with the prefix a number of **BGP attributes**. In BGP jargon, a prefix along with its attributes is called a **route**. Thus, BGP peers advertise routes to each other. Two of the more important attributes are AS-PATH and NEXT-HOP:

- **AS-PATH.** This attribute contains the ASs through which the advertisement for the prefix has passed. When a prefix is passed into an AS, the AS adds its ASN to the AS-PATH attribute. For example, consider Figure 4.40 and suppose that prefix 138.16.64/24 is first advertised from AS2 to AS1; if AS1 then advertises the prefix to AS3, AS-PATH would be AS2 AS1. Routers use the AS-PATH attribute to detect and prevent looping advertisements; specifically, if a router sees that its AS is contained in the path list, it will reject the advertisement. As we'll soon discuss, routers also use the AS-PATH attribute in choosing among multiple paths to the same prefix.
- Providing the critical link between the inter-AS and intra-AS routing protocols, the NEXT-HOP attribute has a subtle but important use. *The NEXT-HOP is the router interface that begins the AS-PATH.* To gain insight into this attribute, let's again refer to Figure 4.40. Consider what happens when the gateway router 3a in AS3 advertises a route to gateway router 1c in AS1 using eBGP. The route includes the advertised prefix, which we'll call x , and an AS-PATH to the prefix. This advertisement also includes the NEXT-HOP, which is the IP address of the router 3a interface that leads to 1c. (Recall that a router has multiple IP addresses, one for each of its interfaces.) Now consider what happens when router 1d learns about this route from iBGP. After learning about this route to x , router 1d may want to forward packets to x along the route, that is, router 1d may want to include the entry (x, l) in its forwarding table, where l is its interface that begins the least-cost path from 1d towards the gateway router 1c. To determine l , 1d provides the IP address in the NEXT-HOP attribute to its intra-AS routing module. Note that the intra-AS routing algorithm has determined the least-cost path to all subnets attached to the routers in AS1, including to the subnet for the link between 1c and 3a. From this least-cost path from 1d to the 1c-3a subnet, 1d determines its router interface l that begins this path and then adds the entry (x, l) to its forwarding table. Whew! In summary, the NEXT-HOP attribute is used by routers to properly configure their forwarding tables.
- Figure 4.41 illustrates another situation where the NEXT-HOP is needed. In this figure, AS1 and AS2 are connected by two peering links. A router in AS1 could learn

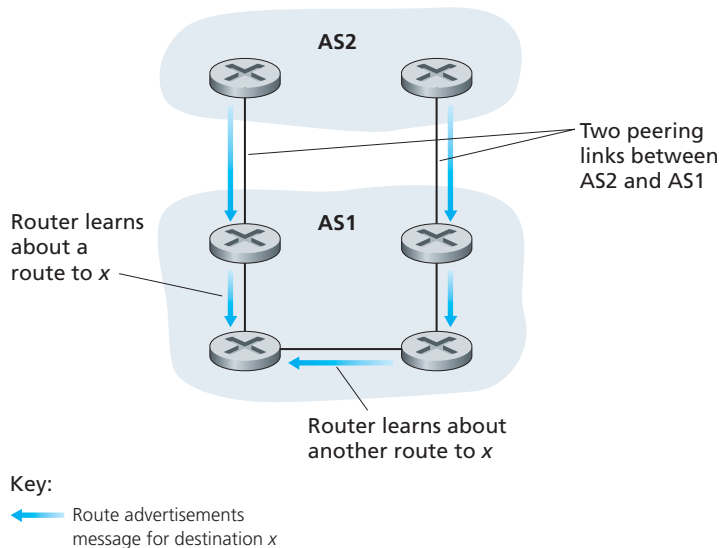


Figure 4.41 ♦ NEXT-HOP attributes in advertisements are used to determine which peering link to use

about two different routes to the same prefix x . These two routes could have the same AS-PATH to x , but could have different NEXT-HOP values corresponding to the different peering links. Using the NEXT-HOP values and the intra-AS routing algorithm, the router can determine the cost of the path to each peering link, and then apply hot-potato routing (see Section 4.5.3) to determine the appropriate interface.

BGP also includes attributes that allow routers to assign preference metrics to the routes, and an attribute that indicates how the prefix was inserted into BGP at the origin AS. For a full discussion of route attributes, see [Griffin 2012; Stewart 1999; Halabi 2000; Feamster 2004; RFC 4271].

When a gateway router receives a route advertisement, it uses its **import policy** to decide whether to accept or filter the route and whether to set certain attributes such as the router preference metrics. The import policy may filter a route because the AS may not want to send traffic over one of the ASs in the route's AS-PATH. The gateway router may also filter a route because it already knows of a preferable route to the same prefix.

BGP Route Selection

As described earlier in this section, BGP uses eBGP and iBGP to distribute routes to all the routers within ASs. From this distribution, a router may learn about more than one route to any one prefix, in which case the router must select one of the

possible routes. The input into this route selection process is the set of all routes that have been learned and accepted by the router. If there are two or more routes to the same prefix, then BGP sequentially invokes the following elimination rules until one route remains:

- Routes are assigned a local preference value as one of their attributes. The local preference of a route could have been set by the router or could have been learned by another router in the same AS. This is a policy decision that is left up to the AS's network administrator. (We will shortly discuss BGP policy issues in some detail.) The routes with the highest local preference values are selected.
- From the remaining routes (all with the same local preference value), the route with the shortest AS-PATH is selected. If this rule were the only rule for route selection, then BGP would be using a DV algorithm for path determination, where the distance metric uses the number of AS hops rather than the number of router hops.
- From the remaining routes (all with the same local preference value and the same AS-PATH length), the route with the closest NEXT-HOP router is selected. Here, closest means the router for which the cost of the least-cost path, determined by the intra-AS algorithm, is the smallest. As discussed in Section 4.5.3, this process is called hot-potato routing.
- If more than one route still remains, the router uses BGP identifiers to select the route; see [Stewart 1999].

The elimination rules are even more complicated than described above. To avoid nightmares about BGP, it's best to learn about BGP selection rules in small doses!



PRINCIPLES IN PRACTICE

PUTTING IT ALL TOGETHER: HOW DOES AN ENTRY GET INTO A ROUTER'S FORWARDING TABLE?

Recall that an entry in a router's forwarding table consists of a prefix (e.g., 138.16.64/22) and a corresponding router output port (e.g., port 7). When a packet arrives to the router, the packet's destination IP address is compared with the prefixes in the forwarding table to find the one with the longest prefix match. The packet is then forwarded (within the router) to the router port associated with that prefix. Let's now summarize how a routing entry (prefix and associated port) gets entered into a forwarding table. This simple exercise will tie together a lot of what we just learned about routing and forwarding. To make things interesting, let's assume that the prefix is a "foreign prefix," that is, it does not belong to the router's AS but to some other AS.

In order for a prefix to get entered into the router's forwarding table, the router has to *first become aware* of the prefix (corresponding to a subnet or an aggregation of subnets). As we have just learned, the router becomes aware of the prefix via a BGP route

advertisement. Such an advertisement may be sent to it over an eBGP session (from a router in another AS) or over an iBGP session (from a router in the same AS).

After the router becomes aware of the prefix, it needs to determine the appropriate output port to which datagrams destined to that prefix will be forwarded, before it can enter that prefix in its forwarding table. If the router receives more than one route advertisement for this prefix, the router uses the BGP route selection process, as described earlier in this subsection, to find the “best” route for the prefix. Suppose such a best route has been selected. As described earlier, the selected route includes a NEXT-HOP attribute, which is the IP address of the first router outside the router’s AS along this best route. As described above, the router then uses its intra-AS routing protocol (typically OSPF) to determine the shortest path to the NEXT-HOP router. The router finally determines the port number to associate with the prefix by identifying the first link along that shortest path. The router can then (finally!) enter the prefix-port pair into its forwarding table! The forwarding table computed by the routing processor (see Figure 4.6) is then pushed to the router’s input port line cards.

Routing Policy

Let’s illustrate some of the basic concepts of BGP routing policy with a simple example. Figure 4.42 shows six interconnected autonomous systems: A, B, C, W, X, and Y. It is important to note that A, B, C, W, X, and Y are ASs, not routers. Let’s assume that autonomous systems W, X, and Y are stub networks and that A, B, and C are backbone provider networks. We’ll also assume that A, B, and C, all peer with each other, and provide full BGP information to their customer networks. All traffic entering a **stub network** must be destined for that network, and all traffic leaving a stub network must have originated in that network. W and Y are clearly stub networks. X is a **multi-homed stub network**, since it is connected to the rest of the network via two different providers (a scenario that is becoming increasingly common in practice). However, like W and Y, X itself must be the source/destination of all traffic leaving/entering X. But how will this stub network behavior be implemented and enforced? How will X be prevented from forwarding traffic between B and C? This can easily be

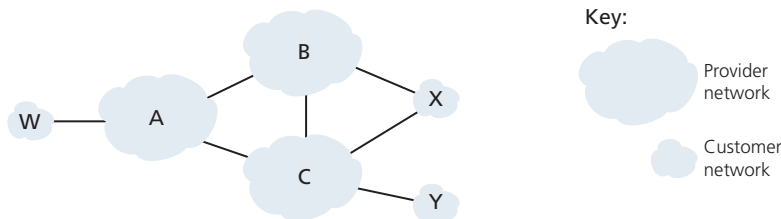


Figure 4.42 ♦ A simple BGP scenario



PRINCIPLES IN PRACTICE

WHY ARE THERE DIFFERENT INTER-AS AND INTRA-AS ROUTING PROTOCOLS?

Having now studied the details of specific inter-AS and intra-AS routing protocols deployed in today's Internet, let's conclude by considering perhaps the most fundamental question we could ask about these protocols in the first place (hopefully, you have been wondering this all along, and have not lost the forest for the trees!): Why are different inter-AS and intra-AS routing protocols used?

The answer to this question gets at the heart of the differences between the goals of routing within an AS and among ASs:

- *Policy.* Among ASs, policy issues dominate. It may well be important that traffic originating in a given AS not be able to pass through another specific AS. Similarly, a given AS may well want to control what transit traffic it carries between other ASs. We have seen that BGP carries path attributes and provides for controlled distribution of routing information so that such policy-based routing decisions can be made. Within an AS, everything is nominally under the same administrative control, and thus policy issues play a much less important role in choosing routes within the AS.
- *Scale.* The ability of a routing algorithm and its data structures to scale to handle routing to/among large numbers of networks is a critical issue in inter-AS routing. Within an AS, scalability is less of a concern. For one thing, if a single administrative domain becomes too large, it is always possible to divide it into two ASs and perform inter-AS routing between the two new ASs. (Recall that OSPF allows such a hierarchy to be built by splitting an AS into areas.)
- *Performance.* Because inter-AS routing is so policy oriented, the quality (for example, performance) of the routes used is often of secondary concern (that is, a longer or more costly route that satisfies certain policy criteria may well be taken over a route that is shorter but does not meet that criteria). Indeed, we saw that among ASs, there is not even the notion of cost (other than AS hop count) associated with routes. Within a single AS, however, such policy concerns are of less importance, allowing routing to focus more on the level of performance realized on a route.

accomplished by controlling the manner in which BGP routes are advertised. In particular, X will function as a stub network if it advertises (to its neighbors B and C) that it has no paths to any other destinations except itself. That is, even though X may know of a path, say XCY, that reaches network Y, it will *not* advertise this path to B. Since B is unaware that X has a path to Y, B would never forward traffic destined to Y (or C) via X. This simple example illustrates how a selective route advertisement policy can be used to implement customer/provider routing relationships.

Let's next focus on a provider network, say AS B. Suppose that B has learned (from A) that A has a path AW to W. B can thus install the route BAW into its routing information base. Clearly, B also wants to advertise the path BAW to its customer, X, so that X knows that it can route to W via B. But should B advertise the path BAW to C? If it does so, then C could route traffic to W via CBAW. If A, B, and C are all backbone providers, then B might rightly feel that it should not have to shoulder the burden (and cost!) of carrying transit traffic between A and C. B might rightly feel that it is A's and C's job (and cost!) to make sure that C can route to/from A's customers via a direct connection between A and C. There are currently no official standards that govern how backbone ISPs route among themselves. However, a rule of thumb followed by commercial ISPs is that any traffic flowing across an ISP's backbone network must have either a source or a destination (or both) in a network that is a customer of that ISP; otherwise the traffic would be getting a free ride on the ISP's network. Individual peering agreements (that would govern questions such as those raised above) are typically negotiated between pairs of ISPs and are often confidential; [Huston 1999a] provides an interesting discussion of peering agreements. For a detailed description of how routing policy reflects commercial relationships among ISPs, see [Gao 2001; Dimitropoulos 2007]. For a discussion of BGP routing policies from an ISP standpoint, see [Caesar 2005b].

As noted above, BGP is the *de facto* standard for inter-AS routing for the public Internet. To see the contents of various BGP routing tables (large!) extracted from routers in tier-1 ISPs, see <http://www.routeviews.org>. BGP routing tables often contain tens of thousands of prefixes and corresponding attributes. Statistics about the size and characteristics of BGP routing tables are presented in [Potaroo 2012].

This completes our brief introduction to BGP. Understanding BGP is important because it plays a central role in the Internet. We encourage you to see the references [Griffin 2012; Stewart 1999; Labovitz 1997; Halabi 2000; Huitema 1998; Gao 2001; Feamster 2004; Caesar 2005b; Li 2007] to learn more about BGP.

4.7 Broadcast and Multicast Routing

Thus far in this chapter, our focus has been on routing protocols that support unicast (i.e., point-to-point) communication, in which a single source node sends a packet to a single destination node. In this section, we turn our attention to broadcast and multicast routing protocols. In **broadcast routing**, the network layer provides a service of delivering a packet sent from a source node to all other nodes in the network; **multicast routing** enables a single source node to send a copy of a packet to a subset of the other network nodes. In Section 4.7.1 we'll consider broadcast routing algorithms and their embodiment in routing protocols. We'll examine multicast routing in Section 4.7.2.